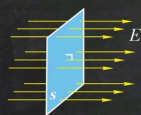


Title:

Date: / /

考点	重要程度	占 分	常见题型
1. 电通量	★★★★	0-3	选择、填空
2. 高斯定理求场强	必 考	5-10	大题

1. 电通量

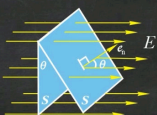


$$\Phi_e = E \cdot S$$

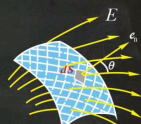
(1) 电通量: $\Phi_e = E \cdot S$

(2) E 和 S 必须是垂直关系

(3) 对于曲面, 向外穿出为正, 向内穿入为负



$$\Phi_e = E \cdot S \cos \theta$$



$$d\Phi_e = \vec{E} \cdot d\vec{S}$$

$$\Phi_e = \int d\Phi_e = \int \vec{E} \cdot d\vec{S}$$

2. 静电场中的高斯定理 $\Phi_e = \oint \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \sum q_{in}$

题1. 求电荷量为 q 的均匀带电球壳内外场强分布。

解: 由 $\oint \vec{E} \cdot d\vec{S} = E \cdot \oint dS = E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} \sum q_{in}$

$$\Rightarrow E = \frac{\sum q_{in}}{4\pi\epsilon_0 r^2}$$

$$\Phi_e = \oint \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \sum q_{in}$$

$$r < R \text{ 时 } \sum q_{in} = 0 \Rightarrow E = 0$$

$$r > R \text{ 时 } \sum q_{in} = q \Rightarrow E = \frac{q}{4\pi\epsilon_0 r^2}$$



$$\text{球的表面积: } S = 4\pi R^2$$

2. 静电场中的高斯定理 $\Phi_e = \oint \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \sum q_{in}$

题2. 半径为 R 的球体, 均匀带电 q , 试求场强分布。

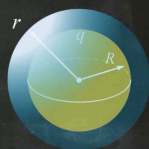
解: 由 $\oint \vec{E} \cdot d\vec{S} = E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} \sum q_{in} \Rightarrow E = \frac{\sum q_{in}}{4\pi\epsilon_0 r^2}$

$$\text{体密度 } \rho = \frac{q}{V} = \frac{q}{\frac{4}{3}\pi R^3} = \frac{3q}{4\pi R^3}$$

$$r < R \text{ 时 } \sum q_{in} = \rho \cdot V_r = \frac{3q}{4\pi R^3} \cdot \frac{4}{3}\pi r^3 = \frac{qr^3}{R^3}$$

$$\text{故 } E = \frac{qr^3/R^3}{4\pi\epsilon_0 r^2} = \frac{qr}{4\pi\epsilon_0 R^3}$$

$$r > R \text{ 时 } \sum q_{in} = q \text{ 故 } E = \frac{q}{4\pi\epsilon_0 r^2}$$



$$\text{球的体积: } V = \frac{4}{3}\pi R^3$$

2. 静电场中的高斯定理 $\Phi_e = \oint \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \sum q_{in}$

题3. 一半径为 R 的带电球体, 其电荷密度 $\rho = A \cdot r$ ($A > 0$), 试求球内外的场强分布。

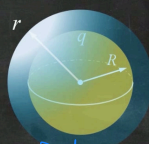
解: 由 $\oint \vec{E} \cdot d\vec{S} = E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} \sum q_{in} \Rightarrow E = \frac{\sum q_{in}}{4\pi\epsilon_0 r^2}$

$$r < R \text{ 时 } \sum q_{in} = \int \rho dV = \int_0^r A \cdot r' \cdot 4\pi r'^2 dr' = \pi A r^4$$

$$\text{故 } E = \frac{\pi A r^4}{4\pi\epsilon_0 r^2} = \frac{A r^2}{4\epsilon_0}$$

$$r > R \text{ 时 } \sum q_{in} = \int \rho dV = \int_0^R A \cdot r' \cdot 4\pi r'^2 dr' = \pi A R^4$$

$$\text{故 } E = \frac{\pi A R^4}{4\pi\epsilon_0 r^2} = \frac{A R^4}{4\epsilon_0 r^2}$$



2. 静电场中的高斯定理 $\Phi_e = \oint \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \sum q_{in}$

题4. 求半径为 R , 线密度为 λ 的无限长均匀带电圆柱体内外场强分布。

解: 由 $\oint \vec{E} \cdot d\vec{S} = E \cdot \oint dS = E \cdot 2\pi r h = \frac{1}{\epsilon_0} \sum q_{in}$

$$\Rightarrow E = \frac{\sum q_{in}}{2\pi\epsilon_0 r h}$$

$$r < R \text{ 时 } \sum q_{in} = \frac{\lambda h}{\pi R^2} \cdot \pi r^2 h = \frac{r^2 \lambda h}{R^2} \Rightarrow E = \frac{r^2 \lambda h}{2\pi\epsilon_0 R^2} = \frac{\lambda r}{2\pi\epsilon_0 R^2}$$

$$r > R \text{ 时 } \sum q_{in} = \lambda h \Rightarrow E = \frac{\lambda h}{2\pi\epsilon_0 r h} = \frac{\lambda}{2\pi\epsilon_0 r}$$

